



interfaceNode - *Manual*

Development 1.0.0

The Atkin Engineering **interfaceNode** is a Wireless RS-485 bridge that provides bridging connections between RS-485, USB, TCP sockets, and WebSockets for various control devices.

front

rear



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1. Connections

Front

- USB Type-C, USB 2.0 full-speed port [^usbpower]
- 1x Status LED
- 1x System button

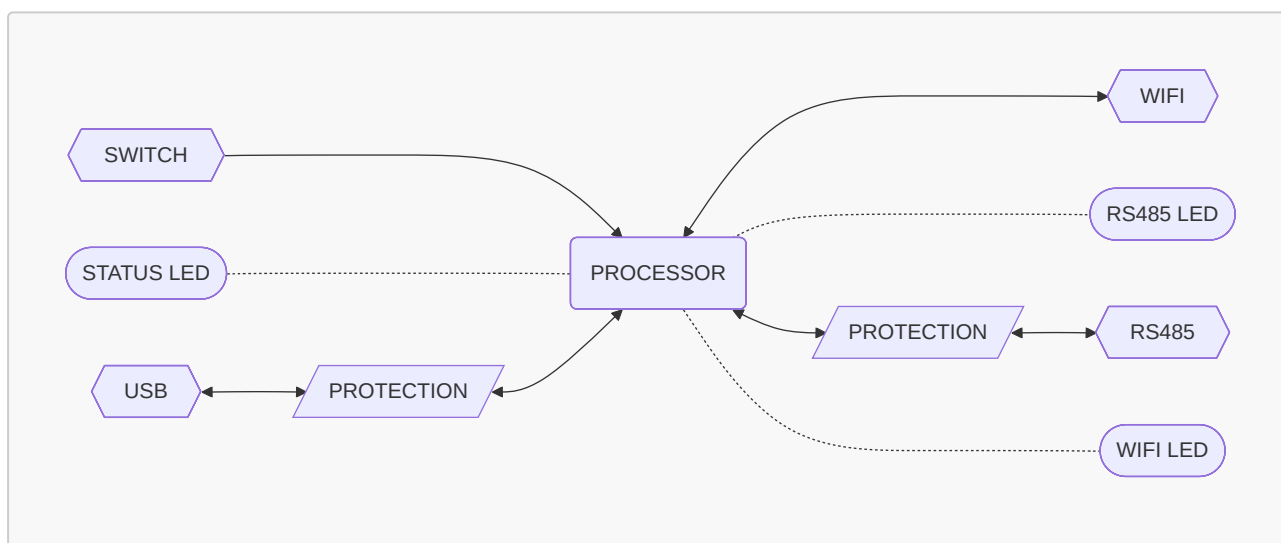
Rear

- 2x RS-485 (parallel RJ12 6P6 ports) supplies power & data [^rs485power]
- 2x RGB status LED's
- Full / Half duplex (hardware jumper - see [^jumpers])
- Termination **None** / **120ohm** (hardware jumper - see [^jumpers])

RJ45 Pinout (RJ12 6P6)

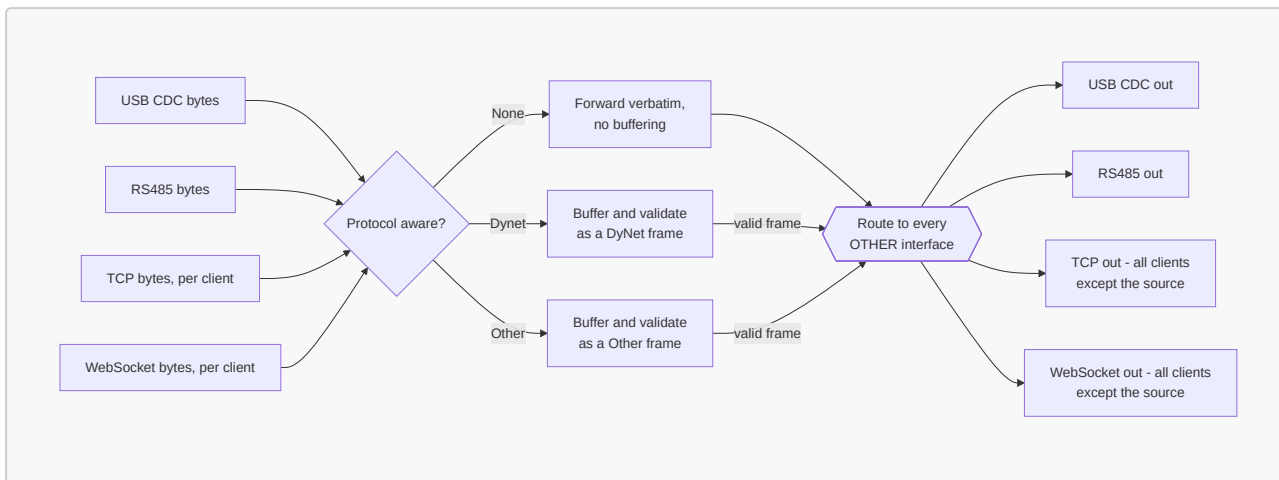
1. Ground
2. Ground
3. Data +
4. Data -
5. 12/24 VDC
6. 12/24 VDC

2. Block diagram



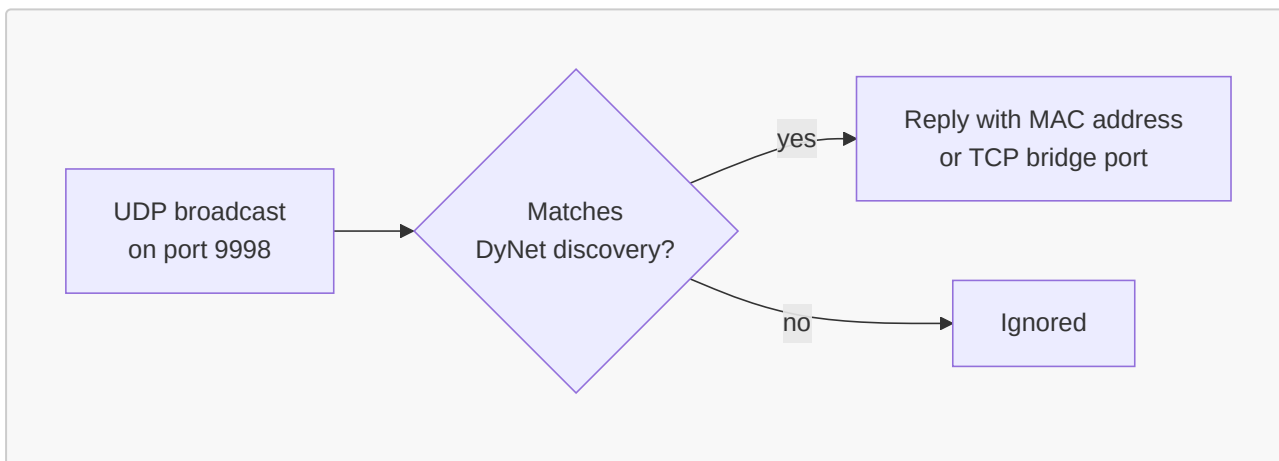
3. Operation flow

Every byte arriving on USB, RS485, a TCP socket, or a WebSocket is handled the same way, regardless of which interface it came in on:



A frame is never sent back out the interface it arrived on. With **Protocol aware** set to **Dynet**, incomplete or checksum-invalid data is held and re-synchronised rather than forwarded; with it set to **None**, every byte is passed straight through with no validation at all.

The DyNet discovery beacon (see [Sockets](#)) is a separate, parallel process - it does not pass through the flow above, and nothing it receives or sends is routed to USB, RS485, or TCP:



4. Default mode

A factory-reset **interfaceNode** boots in **Access Point** mode with:

- SSID: **interfaceNodeXXYY**, where **XXYY** is the last two bytes of the device's MAC address in hex (also its serial number - see [System](#))
- Password: **password**
- IP address: **192.168.4.1**
- DHCP server enabled

5. System button

The system button supports two gestures:

- **Short press** Enables configuration mode and reboots into it - see the [System](#) section for what configuration mode means.

- **2 second hold** Resets all settings to factory defaults and reboots into **default mode**.

Note: the button is not responsive immediately at power-on - it becomes active once the rest of the system (network, bridge or web interface, WebSocket bridge, beacon) has finished starting. It also stops responding once a reboot has been triggered by any means (the button itself, or a web request such as **Save settings**) - a button press during the short delay before the device actually restarts is ignored rather than queued.

6. Config mode

Configuration mode runs the web interface used to view and change every setting on the device - see [System](#) for how it relates to normal bridge operation, and [Default mode](#) / [System button](#) for how to enter it.

The device can also be reached by its mDNS hostname instead of its IP address, e.g. `interfacenode3080.local` (`XXYY` is the same last-two-bytes-of-MAC suffix as the default SSID - see [Default mode](#)). This works on networks that support mDNS/Bonjour, most operating systems out of the box.

The table below lists every setting available while in configuration mode. See the linked section for each group's full explanation - hardware-only settings (RS485 termination and duplex - see [Connections](#)) are not included here, since they are not part of the web interface at all.

System

Setting	Values
Operation mode	Access point / Station - see Operation mode
Protocol aware	None / Dynet - see Operation flow

Access point (see [Operation mode](#))

Setting	Values
SSID	Text
Password	Text, 8+ characters
Channel	1-13
Encryption	Selectable, but currently fixed to WPA2 PSK regardless of selection
Custom IP	Off (default 192.168.4.1) / On (IP address, network mask, gateway, DNS)

Station (see [Operation mode](#))

Setting	Values
SSID	Text
Encryption	Open, WEP, WPA PSK, WPA2 PSK, WPA WPA2 PSK, WPA3 PSK, WPA2 WPA3 PSK, WPA2 Enterprise, WPA3 Enterprise
Password	Text, 8+ characters (non-Enterprise types)
Username / ID	Text (Enterprise types only)
Validate SSL Certificate	Ignore / Validate, plus a certificate upload (Enterprise types only)
DHCP	On, network-assigned (default) / Off (static IP address, network mask, gateway, DNS)

Sockets (see [Sockets](#))

Setting	Values
Port	49152-65535

Serial (see [Serial](#))

Setting	Values
Baud rate	9600 / 19200 / 38400 / 57600 / 115200

System actions (see [System](#))

Setting	Effect
Save settings	Applies and permanently stores every setting above, then reboots
Factory reset	Restores all settings to their factory defaults, saves them, and reboots
Reboot	Restarts the device without changing any settings

7. Operation mode

interfaceNode can operate in two WiFi configurations:

1. **Access point** Creates a WiFi network for other devices to connect to.
2. **Station** Connects to an existing WiFi network as a client.

The **Protocol aware** option controls how the bridge handles data:

- **None** Every byte is forwarded verbatim between interfaces, with no buffering or validation.
- **Dynet** Data is buffered, validated as DyNet 1 or DyNet 2 frames (checksum-checked), and re-synchronised automatically if invalid data is seen.

Access point

- **SSID** SSID for the access point.
- **Password** Password for the access point.
Note: must be at least 8 characters.
- **Channel** WiFi channel (1-13).
- **Encryption** This setting is currently fixed to **WPA2 PSK** regardless of the option selected here - the dropdown does not yet change the device's actual behaviour.

IP address settings:

- **Default IP** **192.168.4.1**, with the built-in DHCP server enabled.
- **Custom IP** Set your own IP address, network mask, gateway, and DNS server. Leaving gateway or DNS blank offers none to connected clients.

Station

- **SSID** SSID of the network to join.
- **Encryption** Security type of the network being joined: **Open**, **WEP**, **WPA PSK**, **WPA2 PSK**, **WPA WPA2 PSK**, **WPA3 PSK**, **WPA2 WPA3 PSK**, **WPA2 Enterprise**, or **WPA3 Enterprise**.
- **Password** Password of the network being joined (WPA/WEP/etc. types). Must be at least 8 characters.

If **WPA2 Enterprise** or **WPA3 Enterprise** is selected, additional fields appear:

- **Username** EAP username.
- **ID** EAP identity string.
- **Ignore SSL Certificate / Validate SSL Certificate** Whether to validate the RADIUS server's certificate.
- **RADIUS/CA Certificate** File picker and **Upload Certificate** button. Upload the RADIUS server's CA certificate (PEM format) here to enable validation.
Note: the certificate is used on the *next* WiFi connection attempt, not immediately - reboot or otherwise trigger a reconnect after uploading. If **Validate SSL Certificate** is selected but no certificate has been uploaded, the connection proceeds without validation.

IP address settings:

- **DHCP** Network-assigned IP address (default).
- **Static** Manually set IP address, network mask, gateway, and DNS server.

8. Sockets

interfaceNode runs a TCP bridge server that any number of clients can connect to over the network, in either Access Point or Station mode.

- **Port** TCP listening port. Must be between 49152 and 65535.

Every TCP client is bridged the same way as USB and RS485 - see [Operation flow](#). Data received from one TCP client is forwarded to USB, RS485, and every *other* connected TCP client, never back to the client that sent it. The number of simultaneous TCP clients is limited by the device's available network sockets, shared with the web configuration interface and other network services.

WebSocket bridge

interfaceNode also runs a dedicated WebSocket server, entirely separate from the TCP bridge above and from the configuration-mode web interface - at `ws://<device address>:8080/ws`. A browser or any WebSocket-capable client connecting here is bridged the same way as USB, RS485, and TCP - see [Operation flow](#).

Two settings, not yet available in the web interface, independently control whether DyNet 1 and DyNet 2 frames are forwarded to WebSocket clients - both are enabled by default. USB, RS485, and TCP routing are unaffected by these settings either way.

DyNet discovery beacon

When **Protocol aware** is set to **Dynet** (see [Operation mode](#)) and the device is running normally (not in configuration mode), **interfaceNode** listens for DyNet discovery broadcasts on UDP port 9998 and replies with its MAC address and TCP bridge port.

Note: this is a separate, independent listener - it does not go through the USB/RS485/TCP bridge described above, and discovery traffic is never forwarded to or from USB, RS485, or TCP clients. See [Operation flow](#) for its own flow, separate from the bridge's.

9. Serial

- **Baud rate** RS485 baud rate: 9600, 19200, 38400, 57600, or 115200.

RS485 duplex mode (full or half) is set via a hardware jumper - see [Connections](#). RS485 is bridged to USB, TCP, and WebSocket clients the same way as any other interface - see [Operation flow](#).

10. Led status

Rebooting (all LEDs, momentarily, immediately before the device restarts):

- **White** Rebooting - shown regardless of what triggered the restart (saving settings, factory reset, a firmware update completing, or the system button).

Boot sequence (all applicable LEDs):

- **Green** Starting up.
- **Yellow** Configuration mode - stays until the device is rebooted out of it.

Once started normally, the status LED shows the active WiFi mode:

- **Magenta** Access Point mode.
- **Blue** Station mode.

RS485 LED

- **Slow red breathe** Idle - no RS485 activity.
- **Fast red blink** RS485 data activity.

WiFi LED

- **Slow blue breathe** No TCP client connected.
- **Solid dim blue** A TCP client is connected, idle.
- **Fast blue blink** TCP socket data activity.

Note: the WiFi LED reflects socket connections and data only - it no longer distinguishes Access Point from Station mode (that's shown on the status LED instead, above).

11. System

- **Save settings** Applies and permanently stores every setting on this page, **then reboots the device**.
- **Factory reset** Restores all settings to their factory defaults, saves them, and reboots.
- **Reboot** Restarts the device without changing any settings.

All three require the **Please confirm** checkbox to be ticked first.

Important: the web configuration interface is only reachable while the device is in configuration mode. During normal bridge operation (USB/RS485/TCP running), the web interface is not available - only the reverse is also true, so switching between configuring the device and using it as a bridge requires a reboot in between.

System log

Recent log output is viewable at </systemlog.html>, colour-coded by severity (error, warning, info, debug, verbose).

Like the rest of the web configuration interface, this page is only reachable while the device is in configuration mode - it can't be checked during normal bridge operation without first rebooting into config mode.

The log is kept in a fixed-size, 128KB in-memory buffer. Once full, it clears itself and starts again from empty rather than growing further or dropping old entries individually.

The log survives a reboot triggered by the device itself (**Save settings**, **Factory reset**, **Reboot**, or the system button) and continues appending rather than starting over. It's only cleared on a genuine power cycle, or after a firmware update - both cases are logged as "STARTING COLD BOOT"; any other restart logs "STARTING WARM BOOT" and keeps everything already there.

12. Firmware

- **Choose file** Choose the firmware bin file from local folder.
- **Upload** Upload firmware to **interfaceNode**. This will cause a reboot, and on occasion will start in **default mode**

Once the browser finishes sending the file, the page shows "Writing to device flash - please wait, do not disconnect..." while the device receives and writes it - this can take several seconds depending on file size. Do not disconnect or reboot during this time.

Firmware files @ <https://github.com/AtkinEngineering/interfaceNode/releases>

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[^usbpower] USB Type-C port does not supply power.

[^rs485power] RS-485 (parallel RJ12 6P6 ports) supplies power & data to **interfaceNode**.

[^jumpers] Termination and duplex are both set via jumpers internal to the enclosure, not configurable from the web interface - changing either requires opening the unit. For duplex specifically: the jumper selects between the receiver always being enabled (full duplex) or tied to the transmit-enable line (half duplex). The firmware drives a single direction-control signal in either case - the jumper, not the firmware, determines which duplex mode results.

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